MSG No. 71.534 $Immm1'$ [Type II, orthorhombic]

Table 1: Wyckoff site: 2a, site symmetry: mmm1'

No.	position	mapping
1	$[0,\ 0,\ 0]$	[1,2,3,4,5,6,7,8,17,18,19,20,21,22,23,24]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[9,10,11,12,13,14,15,16,25,26,27,28,29,30,31,32]

Table 2: Wyckoff site: 2b, site symmetry: mmm1'

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,17,18,19,20,21,22,23,24]
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	[9,10,11,12,13,14,15,16,25,26,27,28,29,30,31,32]

Table 3: Wyckoff site: 2c, site symmetry: mmm1'

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[1,2,3,4,5,6,7,8,17,18,19,20,21,22,23,24]
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	[9,10,11,12,13,14,15,16,25,26,27,28,29,30,31,32]

Table 4: Wyckoff site: 2d, site symmetry: mmm1'

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,17,18,19,20,21,22,23,24]
2	$\left[0,\ \frac{1}{2},\ 0\right]$	[9,10,11,12,13,14,15,16,25,26,27,28,29,30,31,32]

Table 5: Wyckoff site: 4e, site symmetry: 2mm1'

No.	position	mapping
1	$[x,\ 0,\ 0]$	[1,2,7,8,17,18,23,24]
2	$[-x,\ 0,\ 0]$	[3,4,5,6,19,20,21,22]
3	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[9,10,15,16,25,26,31,32]
4	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[11,12,13,14,27,28,29,30]

Table 6: Wyckoff site: 4f, site symmetry: 2mm1'

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	[1,2,7,8,17,18,23,24]
2	$\left[-x,\ \frac{1}{2},\ 0\right]$	[3,4,5,6,19,20,21,22]
3	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[9,10,15,16,25,26,31,32]
4	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	[11,12,13,14,27,28,29,30]

Table 7: Wyckoff site: `4g`, site symmetry: `m2m1'`

No.	position	mapping
1	$[0,\ y,\ 0]$	`[1,3,6,8,17,19,22,24]`
2	$[0,\ -y,\ 0]$	`[2,4,5,7,18,20,21,23]`
3	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[9,11,14,16,25,27,30,32]`
4	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[10,12,13,15,26,28,29,31]`

Table 8: Wyckoff site: `4h`, site symmetry: `m2m1'`

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{2}\right]$	`[1,3,6,8,17,19,22,24]`
2	$\left[0,\ -y,\ \frac{1}{2}\right]$	`[2,4,5,7,18,20,21,23]`
3	$\left[\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$	`[9,11,14,16,25,27,30,32]`
4	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	`[10,12,13,15,26,28,29,31]`

Table 9: Wyckoff site: `4i`, site symmetry: `mm21'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,4,6,7,17,20,22,23]`
2	$[0,\ 0,\ -z]$	`[2,3,5,8,18,19,21,24]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,12,14,15,25,28,30,31]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[10,11,13,16,26,27,29,32]`

Table 10: Wyckoff site: `4j`, site symmetry: `mm21'`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,4,6,7,17,20,22,23]`
2	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[2,3,5,8,18,19,21,24]`
3	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,12,14,15,25,28,30,31]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[10,11,13,16,26,27,29,32]`

Table 11: Wyckoff site: `8k`, site symmetry: `-11'`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,13,17,29]`
2	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[2,14,18,30]`
3	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[3,15,19,31]`
4	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[4,16,20,32]`
5	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[5,9,21,25]`
6	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[6,10,22,26]`
7	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[7,11,23,27]`
8	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[8,12,24,28]`

Table 12: Wyckoff site: `8l`, site symmetry: `m..1'`

No.	position	mapping
1	$[0,\ y,\ z]$	`[1,6,17,22]`
2	$[0,\ -y,\ -z]$	`[2,5,18,21]`
3	$[0,\ y,\ -z]$	`[3,8,19,24]`
4	$[0,\ -y,\ z]$	`[4,7,20,23]`
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,14,25,30]`
6	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[10,13,26,29]`
7	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[11,16,27,32]`
8	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[12,15,28,31]`

Table 13: Wyckoff site: `8m`, site symmetry: `.m.1'`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,7,17,23]`
2	$[x,\ 0,\ -z]$	`[2,8,18,24]`
3	$[-x,\ 0,\ -z]$	`[3,5,19,21]`
4	$[-x,\ 0,\ z]$	`[4,6,20,22]`
5	$\left[x+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,15,25,31]`
6	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[10,16,26,32]`
7	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[11,13,27,29]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[12,14,28,30]`

Table 14: Wyckoff site: `8n`, site symmetry: `..m1'`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,8,17,24]`
2	$[x,\ -y,\ 0]$	`[2,7,18,23]`
3	$[-x,\ y,\ 0]$	`[3,6,19,22]`
4	$[-x,\ -y,\ 0]$	`[4,5,20,21]`
5	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[9,16,25,32]`
6	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[10,15,26,31]`
7	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[11,14,27,30]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[12,13,28,29]`

Table 15: Wyckoff site: `16o`, site symmetry: `11'`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1,17]`
2	$[x,\ -y,\ -z]$	`[2,18]`
3	$[-x,\ y,\ -z]$	`[3,19]`
4	$[-x,\ -y,\ z]$	`[4,20]`
5	$[-x,\ -y,\ -z]$	`[5,21]`
6	$[-x,\ y,\ z]$	`[6,22]`
7	$[x,\ -y,\ z]$	`[7,23]`

continued ...

Table 15

No.	position	mapping
8	$[x, y, -z]$	[8,24]
9	$\left[x+\frac{1}{2}, y+\frac{1}{2}, z+\frac{1}{2}\right]$	[9,25]
10	$\left[x+\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}-z\right]$	[10,26]
11	$\left[\frac{1}{2}-x, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[11,27]
12	$\left[\frac{1}{2}-x, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[12,28]
13	$\left[\frac{1}{2}-x, \frac{1}{2}-y, \frac{1}{2}-z\right]$	[13,29]
14	$\left[\frac{1}{2}-x, y+\frac{1}{2}, z+\frac{1}{2}\right]$	[14,30]
15	$\left[x+\frac{1}{2}, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[15,31]
16	$\left[x+\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[16,32]